

Shreyas Gupta

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PROFESSIONAL SUMMARY

AI Researcher and Machine Learning Engineer pursuing Integrated M.Tech in Artificial Intelligence. Hands-on expertise in generative AI, agentic AI systems, multi-agent architectures, RAG pipelines, transformer-based models, and graph AI. Proven ability to design novel AI algorithms, build production-grade AI systems, and collaborate across cross-functional research teams. Winner of Agentic AI Hackathon 2025 (NielsenIQ & TCS). Passionate about advancing research in neural architectures, knowledge representation, causal reasoning, and autonomous AI systems.

EDUCATION

SRM Institute of Science and Technology

Chennai, India

Integrated M.Tech in Artificial Intelligence (CGPA: 8.29/10)

Aug 2022 – May 2027

- Coursework: Machine Learning, Deep Learning, Neural Network Architectures, Statistical Learning Theory, NLP, Computer Vision, Graph Neural Networks, Reinforcement Learning, AI Agents & Multi-Agent Systems

EXPERIENCE

AI/ML Engineer Intern

May 2026 – July 2026

Revenuxe

Bangalore, India

- Designed and implemented RAG (Retrieval-Augmented Generation) pipelines and agentic AI architectures with multi-agent orchestration using Python, FastAPI, and LangChain across two enterprise AI products.
- Researched and developed novel prompt engineering strategies and attention-based retrieval mechanisms to improve generative AI model performance on domain-specific benchmarks.
- Architected autonomous AI agent systems with tool-use capabilities, chain-of-thought reasoning, and knowledge representation modules for complex enterprise workflows.
- Shipped full-stack AI applications, migrated inference backends to AWS, and Dockerized deployments for scalable model serving.

Cybersecurity Research Intern

Dec 2025 – March 2026

Indian Institute of Technology, Guwahati

Guwahati, India

- Conducted research on graph-based AI models for cybersecurity vulnerability detection, designing modular architecture with CLI, parser, validator, loader, queries, and UI modules.
- Implemented custom LadyBugDB embedded graph database with novel graph traversal algorithms for representing function call graphs and control flow edges – applicable to Graph AI and knowledge representation.
- Applied causal reasoning techniques to identify vulnerability propagation paths in software dependency graphs, contributing to explainable AI for security analysis.

Open Source Contributor (AI Engineer)

April 2023 – Present

NgKore Foundation

Remote

- Built autonomous Python AI agents leveraging LLM-based reasoning for monitoring and configuration management of 5G core network – demonstrating agentic and multi-agentic systems.
- Contributed code fixes and features via GitHub PRs, reviewed and merged by senior maintainers in international open-source AI community.

Business Strategy Intern

May 2024 – July 2024

To-Let Globe

Remote

- Analyzed rental datasets of 10,000+ listings using SQL and delivered data-driven insights via Power BI dashboards for leadership decision-making.

RESEARCH PROJECTS

BharatGreen AI | Python, FastAPI, Next.js, TypeScript, NVIDIA NIM

Jan 2026 – March 2026

- Built AI sustainability platform integrating generative AI models (NVIDIA NIM) for energy forecasting, carbon analysis, and water optimization across Indian regions.
- Implemented what-if analysis engine using causal inference and temporal modeling for carbon-aware scheduling, advancing AI for UN SDGs.
- Developed multimodal data pipeline handling temporal, geospatial, and text data using transformer-based architectures.

Light Pollution Sentinel | Python, Next.js, PostgreSQL, OpenAI, Docker

July 2025 – Sept 2025

- Built autonomous AI agent pipeline detecting light-pollution hotspots from NASA satellite data across 742 districts using concept discovery and extraction.
- Developed explainable AI module generating severity-based alerts with natural language explanations.
- Shipped production system with real-time heatmaps, AI-generated insights, and automated alerting.

AWARDS & ACHIEVEMENTS

Winner – Agentic AI Hackathon 2025 (Certificate) <i>NielsenIQ & Tata Consultancy Services (TCS) – ALPS (Live)</i>	Sep 2025 Chennai, India
Member – Indian Society for Technical Education (ISTE) (Certificate) <i>SRM Institute of Science and Technology (SM No. 2258)</i>	Mar 2025 – Present Chennai, India

TECHNICAL SKILLS

AI/ML Frameworks: PyTorch, TensorFlow, LangChain, OpenAI API, Google Gemini API, NVIDIA NIM, Hugging Face Transformers, scikit-learn, NLTK, Seaborn, Streamlit
AI Research Areas: Generative AI (LLMs), Agentic AI, Multi-Agent Systems, RAG, Transformer Architectures, Attention Mechanisms, Graph AI, Explainable AI, Causal Reasoning, Knowledge Representation
Languages: Python, Java, C/C++, SQL (PostgreSQL), JavaScript, TypeScript
Frameworks: FastAPI, React, Next.js, Node.js, Flask
Developer Tools: AWS, Azure, Docker, Linux, Git, GitHub, VS Code, Jupyter Notebook, LaTeX

CERTIFICATIONS

TATA GenAI Powered Data Analytics Job Simulation , Forage (Certificate)	Oct '25
Deloitte Technology Job Simulation , Forage (Certificate)	Jun '25
Natural Language Processing , NPTEL (Certificate)	Apr '25
Competitive Programming (Certificate of Excellence) , Coding Ninjas (Certificate)	Jun '24
Programming in Java , NPTEL (Certificate)	Oct '23